Learning to Reason under Off-Policy Guidance

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October 27, 2025

Outline

The Problem with On-Policy Learning

- Reinforcement Learning with Verifiable Rewards (RLVR) has shown success in training Large Reasoning Models (LRMs).
- However, these methods are inherently **on-policy**.
- **Limitation**: Learning is restricted to the model's own outputs. The model can only amplify existing behaviors and struggles to acquire reasoning abilities beyond its initial capabilities.
- **Core Question**: How can we empower models to learn reasoning behaviors that surpass their initial cognitive boundaries?

Learning to Reason Under oFF-policY guidance

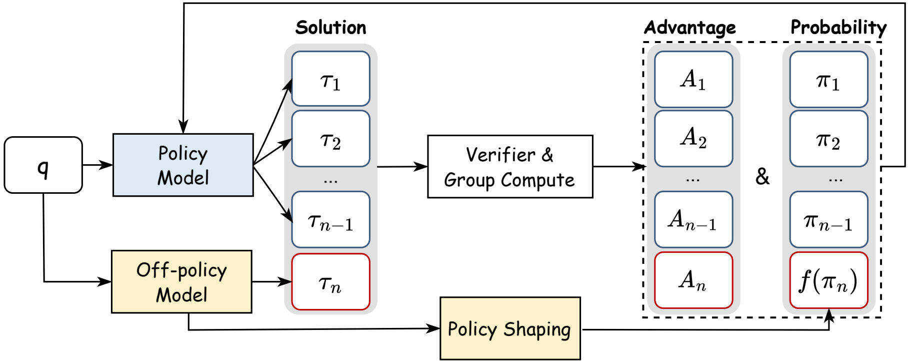

Figure: LUFFY integrates off-policy reasoning traces into RL by combining them with on-policy rollouts, using policy shaping to balance imitation and exploration.

Mixed-Policy GRPO: Core Idea

- LUFFY extends Group Relative Policy Optimization (GRPO) by incorporating **off-policy guidance** (e.g., reasoning traces from a stronger model).
- It combines on-policy rollouts ($\mathcal{G}_{\text{on}}$) with off-policy demonstrations ($\mathcal{G}_{\text{off}}$) to compute the advantage $\hat{A}_i$.

$$\hat{A}_i = \frac{R(\tau_i) - \text{mean}(\mathcal{G}_{\text{on}} \cup \mathcal{G}_{\text{off}})}{\text{std}(\mathcal{G}_{\text{on}} \cup \mathcal{G}_{\text{off}})}$$

- **Intuition**: The model imitates high-quality off-policy traces when its own attempts fail, but explores on its own when it succeeds. This creates a dynamic balance between imitation and exploration.

Mixed-Policy GRPO: The Objective Function

The objective function includes terms for both off-policy and on-policy trajectories.

$$\mathcal{J}_{\text{Mixed}}(\theta) = \frac{1}{Z}(\underbrace{\sum_{j=1}^{N_{\text{off}}} \sum_{t=1}^{|\tau_j|} \text{CLIP}(\hat{r}_{j,t}(\theta, \phi), \hat{A}_j, \epsilon)}_{\text{off-policy objective}} + \underbrace{\sum_{i=1}^{N_{\text{on}}} \sum_{t=1}^{|\tau_i|} \text{CLIP}(r_{i,t}(\theta), \hat{A}_i, \epsilon)}_{\text{on-policy objective}})$$

- Importance sampling ratios are used to correct the policy gradient:
  - On-policy: $r_{i,t}(\theta) = \frac{\pi_\theta(\tau_{i,t}|\dots)}{\pi_{\theta_{\text{old}}}(\tau_{i,t}|\dots)}$
  - Off-policy: $\hat{r}_{j,t}(\theta, \phi) = \frac{\pi_\theta(\tau_{j,t}|\dots)}{\pi_\phi(\tau_{j,t}|\dots)}$

The Challenge: Entropy Collapse

- While Mixed-Policy GRPO works, it can lead to a new problem: **Entropy Collapse**.
- The model converges too quickly, leading to deterministic outputs and reduced exploration.
- This happens because the model tends to reinforce off-policy tokens that are already likely under its own policy, ignoring low-probability (but potentially crucial) tokens that represent new reasoning steps.

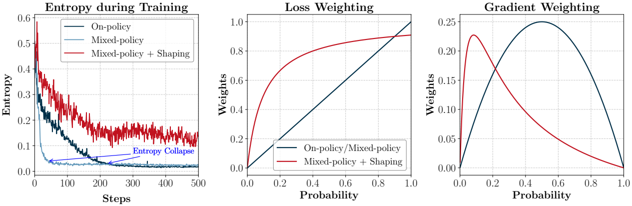

Figure: Left: Entropy collapses faster in Mixed-Policy GRPO compared to On-Policy RL.

Policy Shaping via Regularized Importance Sampling

- To counter entropy collapse, LUFFY introduces **policy shaping**.
- The idea is to re-weight the gradient of the off-policy distribution to amplify the learning signal for low-probability tokens.
- This is done by replacing the off-policy importance sampling ratio $\hat{r}_{j,t}$ with a transformation $f(\hat{r}_{j,t})$.

$$\mathcal{J}_{\text{SHAPING-OFF}}(\theta) \propto \sum_{j=1}^{N_{\text{off}}} \sum_{t=1}^{|\tau_j|} f(\hat{r}_{j,t}(\theta, \phi)) \cdot \hat{A}_j$$

- The paper uses the shaping function $f(x) = \frac{x}{x+\gamma}$, which reweights the gradients to assign more importance to low-probability actions.

How Policy Shaping Works

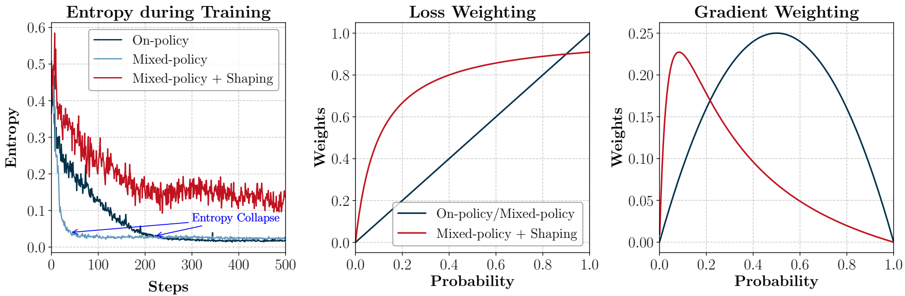

Figure: Middle: The shaping function $f(x)$ value. Right: The resulting gradient weights are amplified for low-probability actions.

- The derivative of the shaping function, $f'(\pi_\theta)$, acts as a weight on the gradient.
- As shown in the figure, this weighting scheme ensures that even when an action's probability π_θ is very low, its gradient receives significant weight.